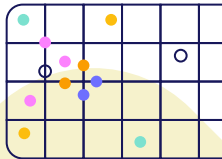
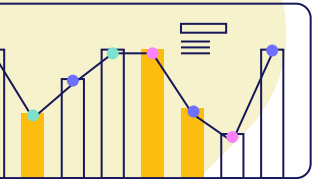


Statistical Analysis for Fungal Growth

By: Lydia Niu



Outline

01.

Background

02.

EDA

03.

Statistical Model

04.

Data Analysis

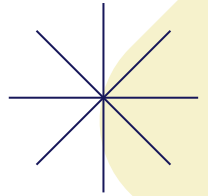
- ANOVA Table
- Assumption Check

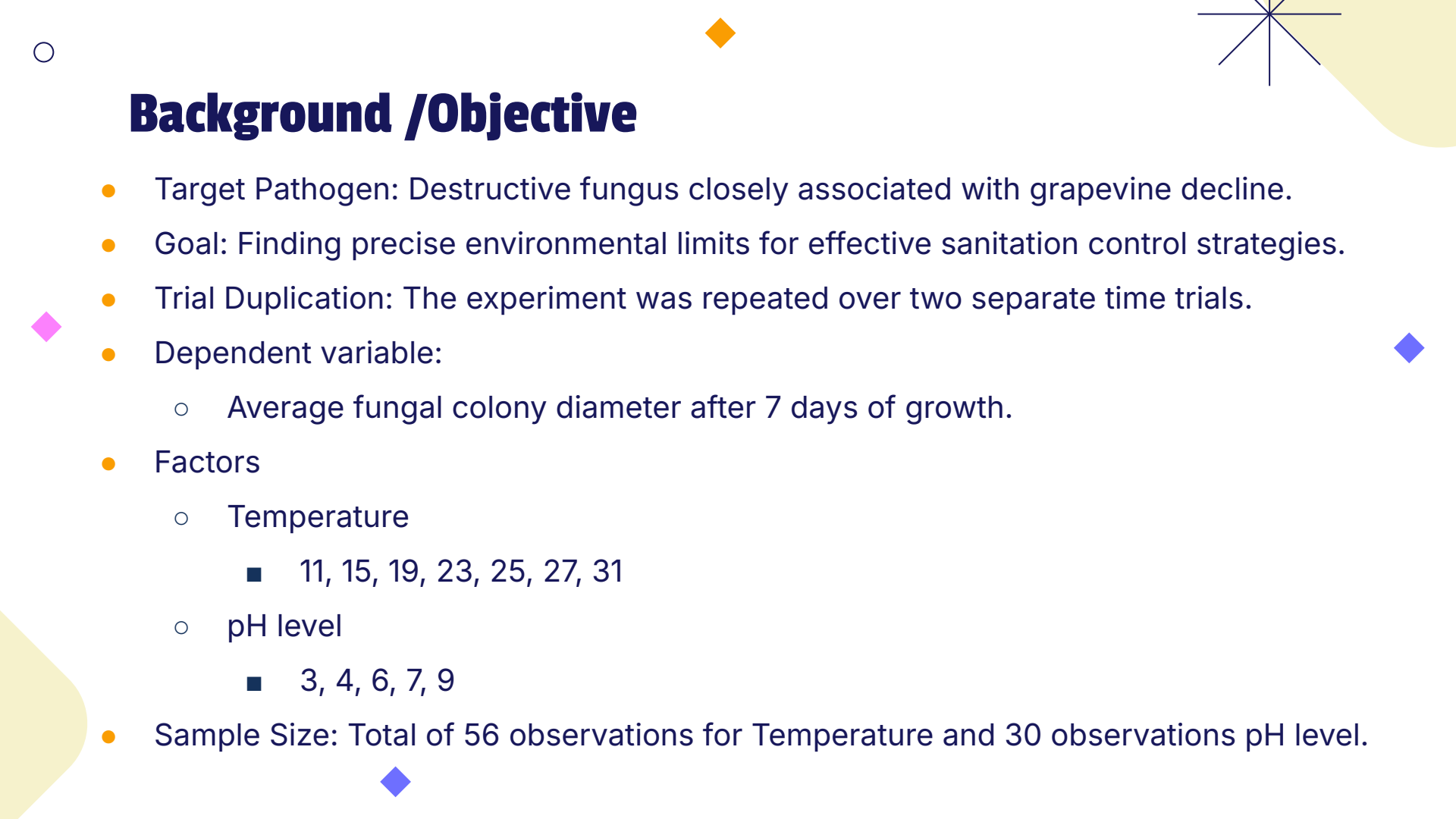
05.

Linear Regression

06.

Conclusion



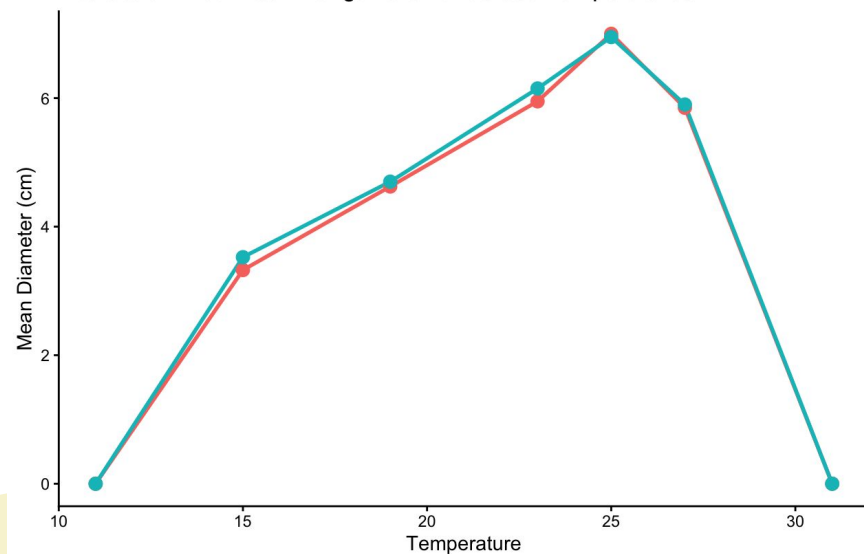


Background /Objective

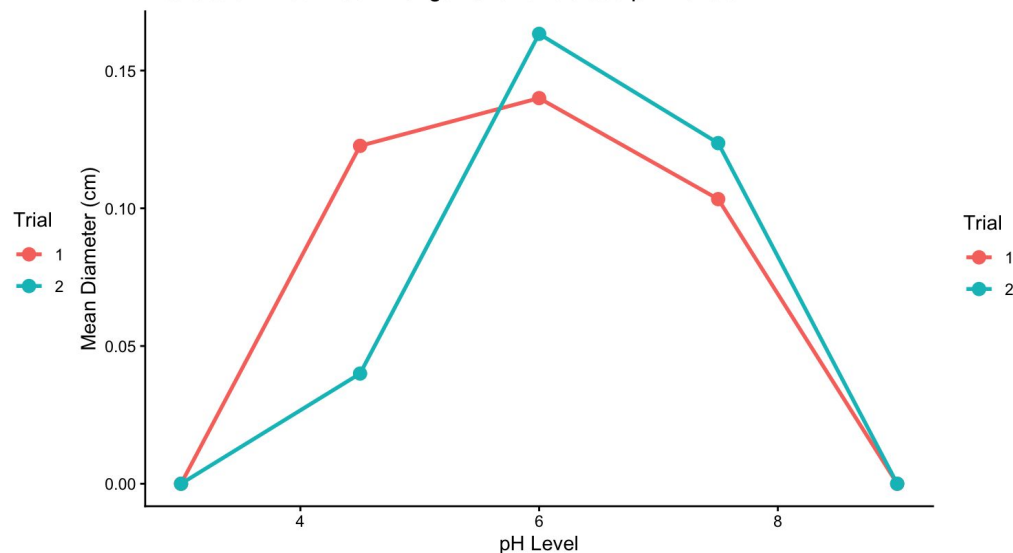
- Target Pathogen: Destructive fungus closely associated with grapevine decline.
- Goal: Finding precise environmental limits for effective sanitation control strategies.
- Trial Duplication: The experiment was repeated over two separate time trials.
- Dependent variable:
 - Average fungal colony diameter after 7 days of growth.
- Factors
 - Temperature
 - 11, 15, 19, 23, 25, 27, 31
 - pH level
 - 3, 4, 6, 7, 9
- Sample Size: Total of 56 observations for Temperature and 30 observations pH level.

EDA - Interaction Plot Across Trials

Interaction Plot: Mean Fungal Growth across Temperatures



Interaction Plot: Mean Fungal Growth across pH Levels

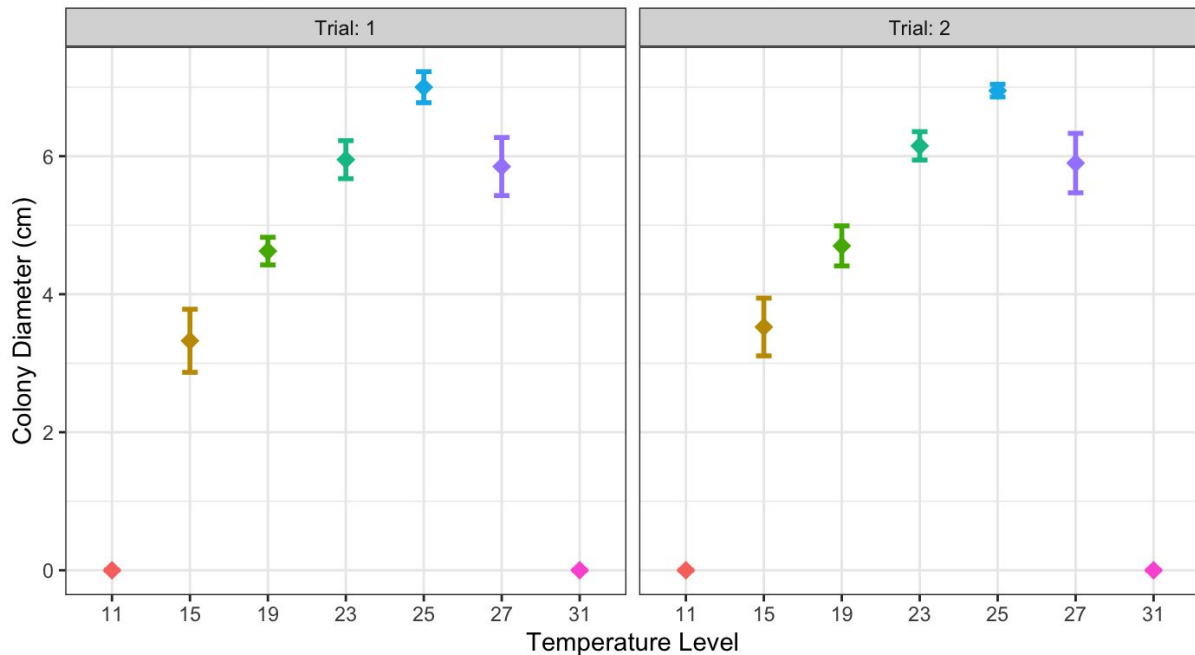


The temperature experiment shows high reproducibility, as the Trial factor has no significant impact on fungal growth across different runs. Conversely, the pH experiment displays significant run-to-run variation, necessitating the inclusion of the Trial interaction to accurately capture and model the sensitive biological response.

EDA - Mean Fungal Growth Across Temperature

Mean Fungal Growth (95% CI) Across Temperature Treatments

Separated by Experimental Trial to view distribution consistency

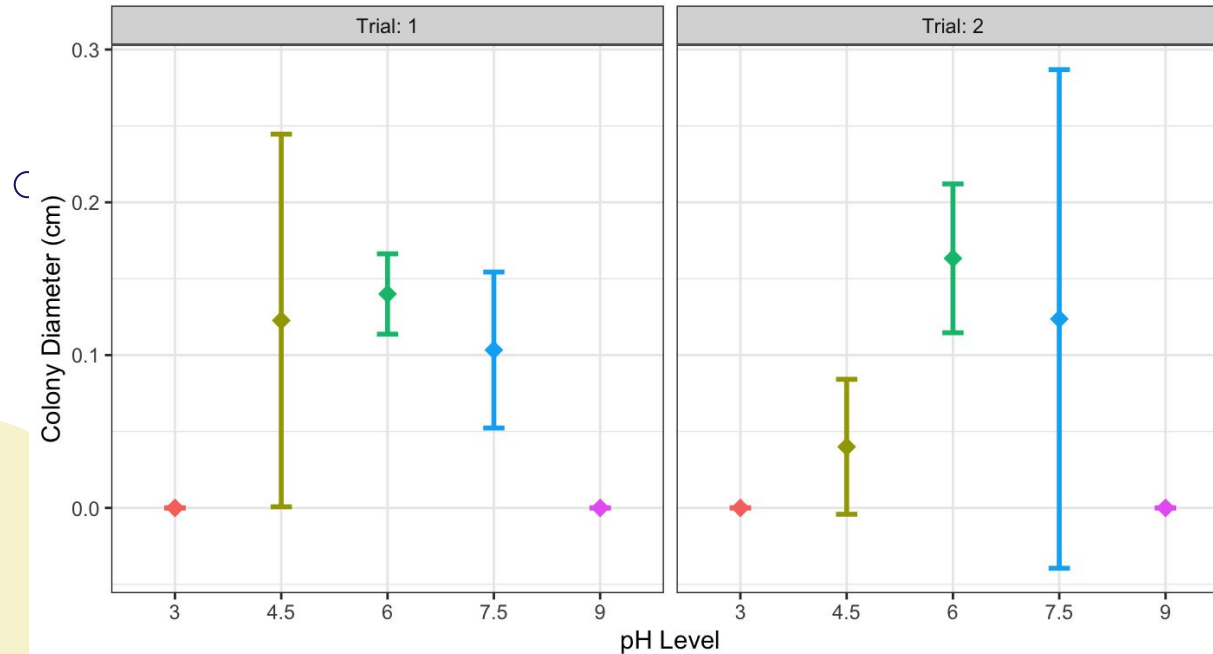


The graphs show a very identical, bell-shaped curve with the peak at temperature 25 and slight deviation for each temperature level. Trial 1 or 2 has no effect on the temperature treatment, suggest that temperature is a strong correlated factor for the fungal growth.

Note that at temperatures 11 and 31, there are no growth, suggest that the type of fungus we are experimenting with could not live under 15 degree and over 31 degree.

EDA - Mean Fungal Growth Across pH level

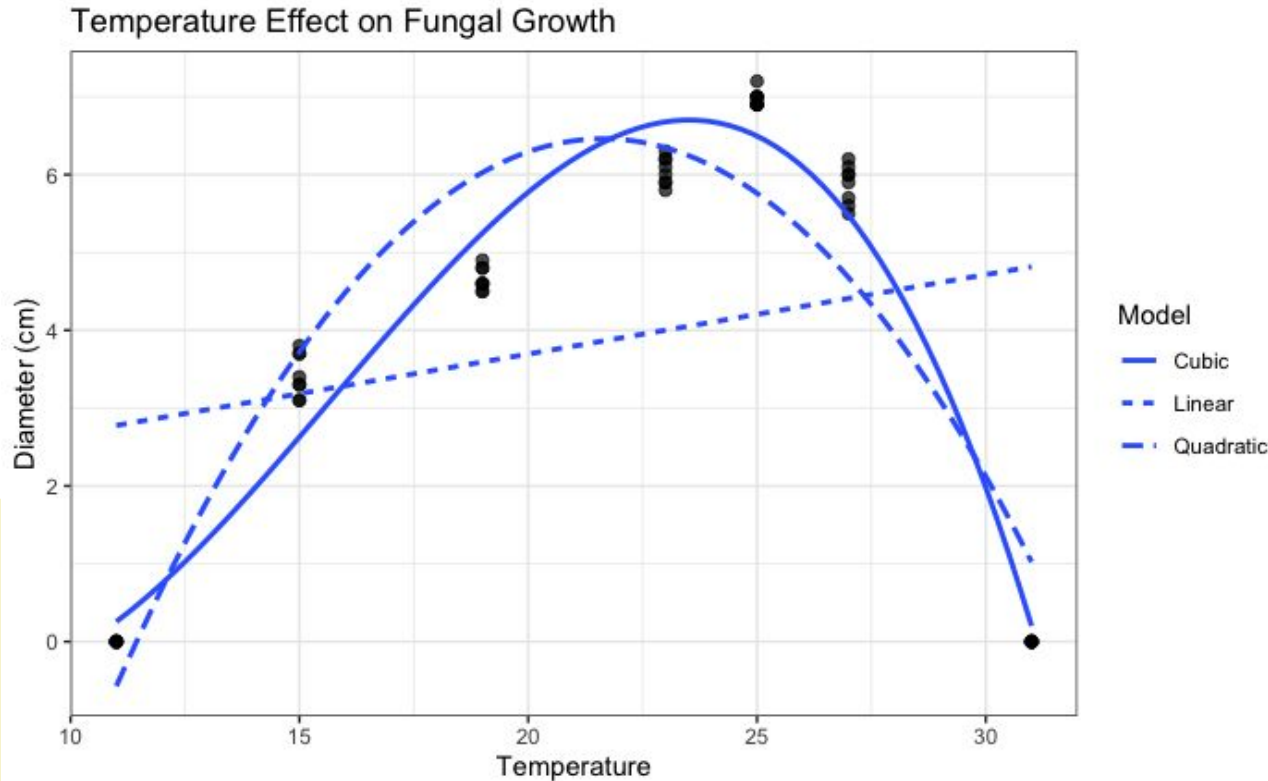
Mean Fungal Growth (95% CI) Across pH Treatments
Separated by Experimental Trial to highlight interaction changes



This graph shows that even under the same pH level environment, the deviation and variance of the colony diameter could also be highly spread out. Suggest that there are other factors to include for data explanation.

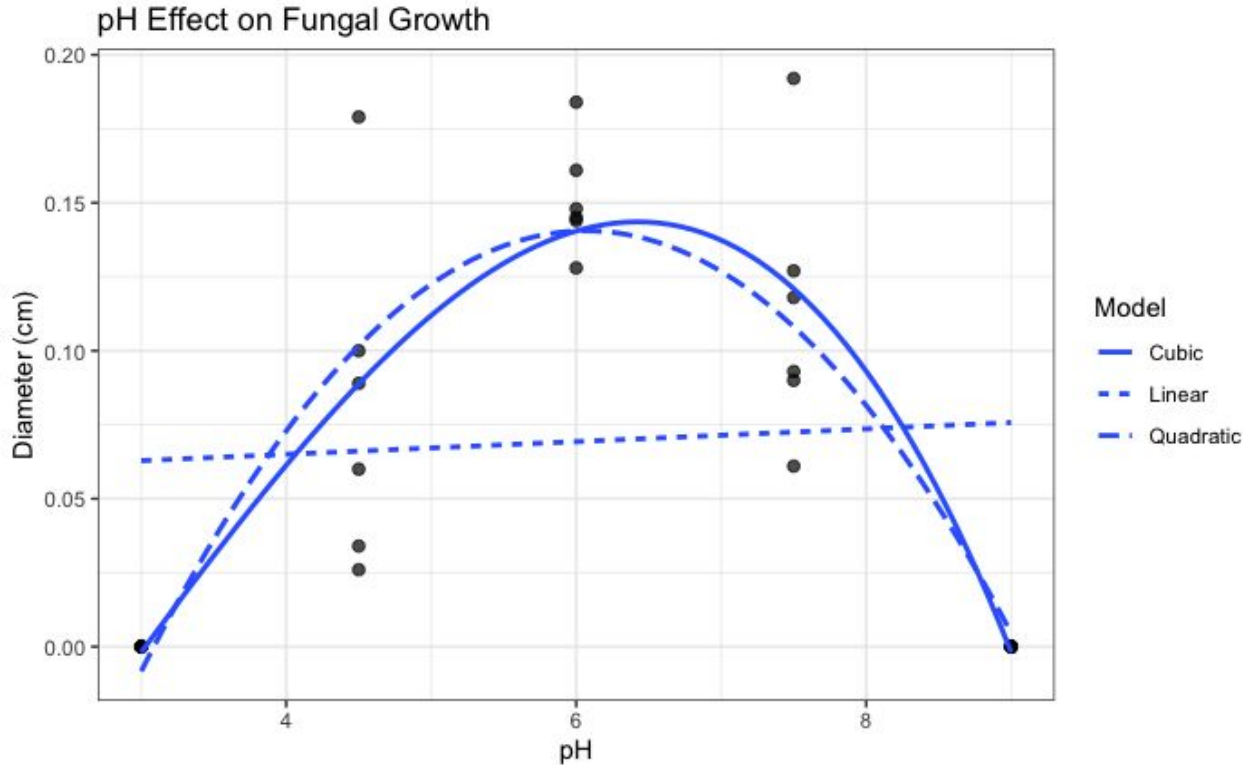
In addition, no results in pH level 3 and 9, suggest that fungus could not grow under high acidic or base condition.

EDA - Comparison of Polynomial Terms of Temperature



From the plot, the quadratic temperature effect provides the best fit to the observed data. This suggests that the nonlinear relationship between temperature and the response is important and should be retained in modeling.

EDA - Comparison of Polynomial Terms of pH Level



The quadratic and cubic curves are nearly identical, suggesting that fungal growth reaches its optimum at a pH between 6 and 7.

This also indicates that polynomial terms are important for capturing the nonlinear effect of pH on growth.

Statistical Model (ANOVA Model)

$$Y_{ijk} = \mu + \tau_i + \beta_j + \tau\beta_{ij} + \epsilon_{ijk}$$

○ *i* - treatment scale
(temperature/ pH level)

j - trial

k - replicates

$$\epsilon \sim N(0, \sigma^2)$$

Where:

Y_{ijk} : response measurement

μ : overall mean

τ_i : effect of treatments (temperature/ pH level)

β_j : effect of trials

$\tau\beta_{ij}$: interaction effect

ϵ_{ijk} : random error

Why Exclude Zero-Diameter Data Points?

- **Extreme Physiological Limits:**
 - Conditions of extreme temperature (11 °C and 31 °C) or pH (3 and 9)
 - Fungal growth is biologically inhibited.
- **Removal of Non-Growth Artifacts:**
 - Do not represent a measurable growth rate → boundary of survival
 - Negatively affected the model performance and deflate the intercept.
- **Impact on Model Complexity:**
 - A critical consequence of this filtering is that the pH dataset was reduced to only three unique levels (4.5, 6.0, 7.5), mathematically preventing the estimation of a cubic (third-degree) term due to lack of degrees of freedom.

Why Include Higher Order Terms?

Model	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1: Linear	38	13.2402				
2: Quadratic	37	8.1632	1	5.077	54.313	<0.001
3: Cubic	36	3.3652	1	4.798	51.328	<0.001

Table 1: ANOVA Model Comparison for Temperature (Linear vs. Quadratic vs. Cubic)

Model	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1: Linear	16	0.03938				
2: Quadratic	15	0.02761	1	0.01177	6.3965	<0.05

Table 2: ANOVA Model Comparison for pH (Linear vs. Quadratic)

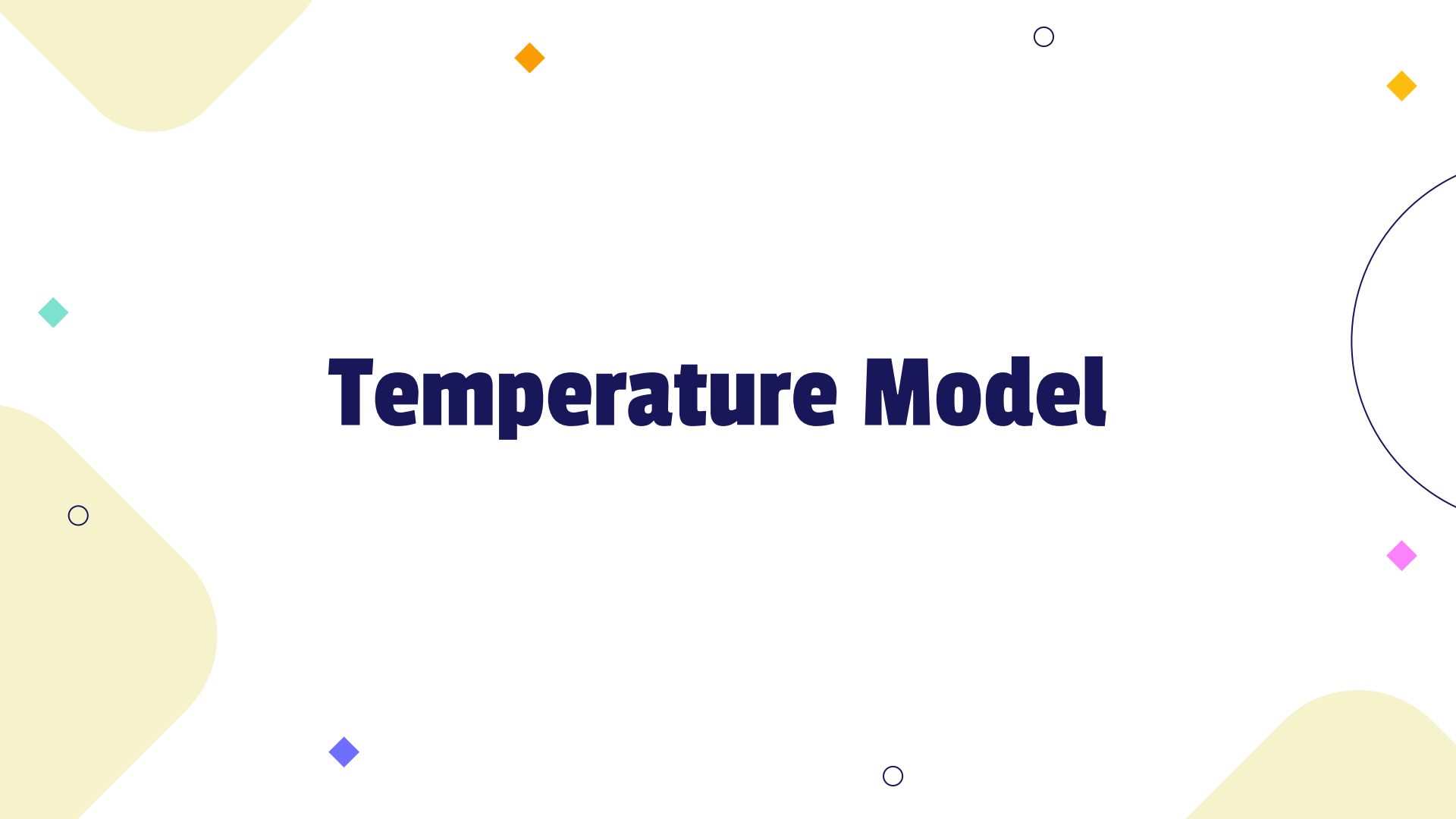
Temperature:

The high F-statistic (51.33) indicates that the cubic term significantly improves the model, showing that the temperature response is nonlinear.

pH:

The significant F-statistic (6.39) supports using a quadratic term, suggesting that fungal growth peaks at an optimal pH rather than changing linearly.

Temperature Model

The background is white with several decorative elements: a large yellow shape in the top-left corner, a yellow shape in the bottom-left corner, and a yellow shape in the bottom-right corner. There are also several small geometric shapes scattered around: an orange diamond at the top, a teal diamond on the left, a blue diamond at the bottom, a yellow diamond at the top-right, and a pink diamond at the bottom-right. Additionally, there are several small circles: a blue circle at the top, a blue circle on the left, a blue circle at the bottom, and a blue circle on the right. A thin blue line forms a partial circle on the right side.

ANOVA table

Predictor	Df	Sum Sq	Mean Sq	F value	p-value
Temp	1	48.770	48.770	481.615	<0.001
Temp ²	1	5.077	5.077	50.137	<0.001
Temp ³	1	4.798	4.798	47.382	<0.001
Trial	1	0.090	0.090	0.891	0.352
Interactions					
Temp:Trial	1	0.033	0.033	0.322	0.574
Temp ² :Trial	1	0.000	0.000	0.000	0.999
Temp ³ :Trial	1	0.002	0.002	0.019	0.891
Residuals	32	3.240	0.101		

Table 3: ANOVA Table for Temperature Regression

Response Variable: Diameter

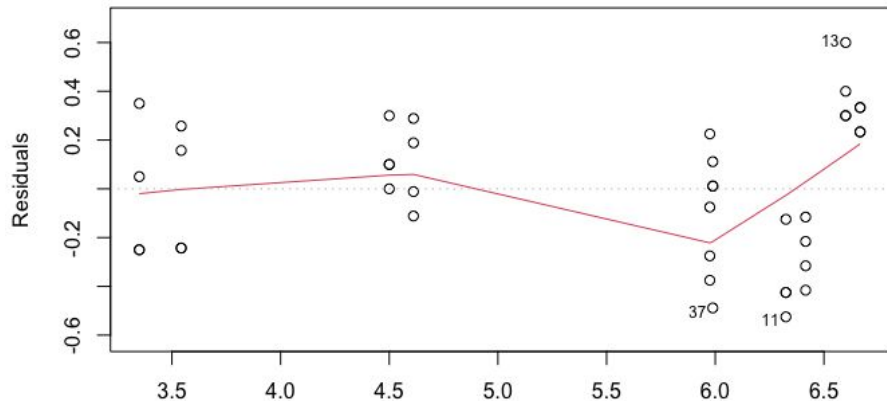
* Significance level: 0.05

- All the polynomial terms of Temperature are extremely significant ($p < 0.001$).
- Trial and all Interaction terms are not significant ($p > 0.05$).
- **Conclusion:** Drop the Trial and the interaction terms for simplicity, keep the curvature terms.

Assumption Checks: Linearity, Homoscedasticity

Residuals vs Fitted: Temperature Model

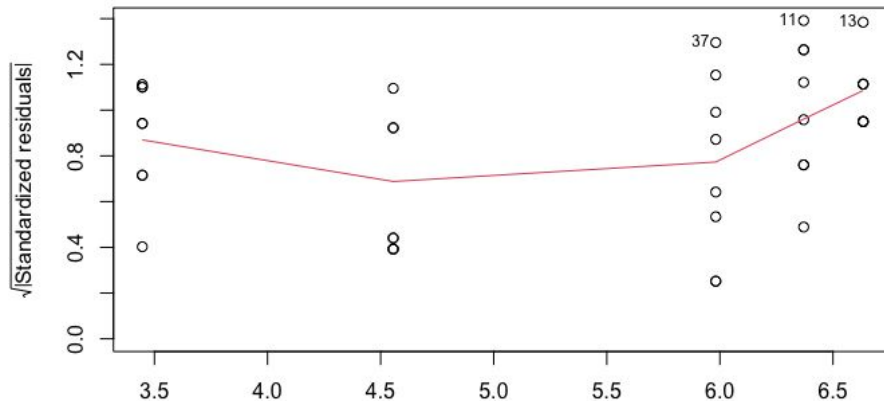
Residuals vs Fitted



Fitted values
 $\text{lm}(\text{Diameter} \sim (\text{Temp} + \text{l}(\text{Temp}^2) + \text{l}(\text{Temp}^3)) * \text{Trial})$

Scale-Location: Temperature Model

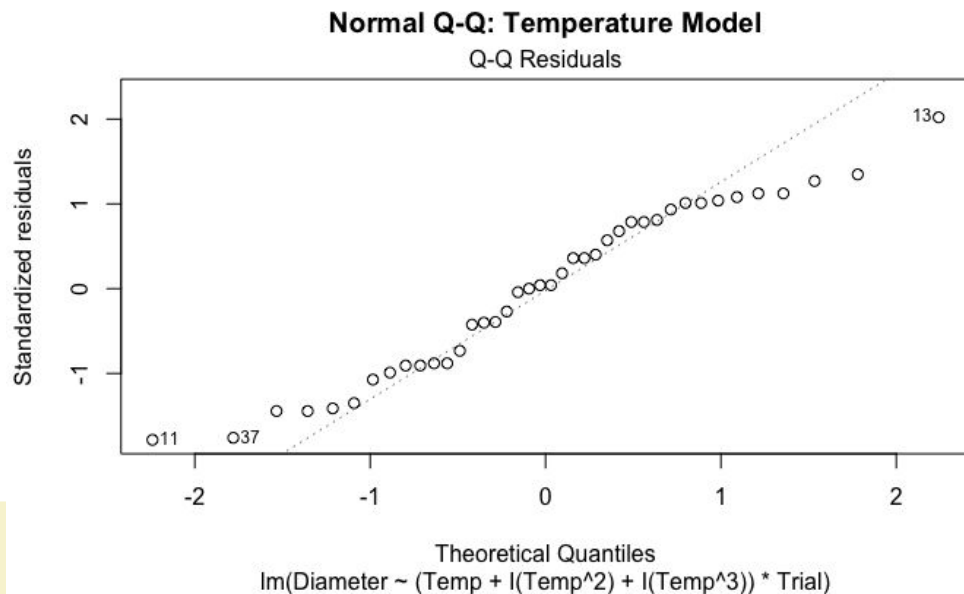
Scale-Location



Fitted values
 $\text{lm}(\text{Diameter} \sim (\text{Temp} + \text{l}(\text{Temp}^2) + \text{l}(\text{Temp}^3)))$

The Residuals vs Fitted plot shows a mostly random scatter around the zero line, suggesting the cubic model successfully captures the non-linear trend. The Scale-Location plot shows a relatively horizontal red line and a consistent spread of points, indicating that the assumption of **constant variance (homoscedasticity)** is satisfied.

Assumption Checks: Normality



Shapiro-Wilk normality test

$$W = 0.95472$$
$$p\text{-value} = 0.1102$$

The null hypothesis states that the residuals are normally distributed. With a p-value of 0.1102, we fail to reject null hypothesis.

The Normal Q-Q plot shows residuals following the diagonal reference line with only minor deviations at the extreme tails. This visual evidence is supported by the Shapiro-Wilk test ($p=0.1102>0.05$), which formally confirm satisfaction of Normality assumption.

Assumption Checks: Independence

Durbin-Watson independence test

$$\begin{array}{|l} \text{D-W stat} = 1.44 \\ \text{p-value} = 0.018 \end{array}$$

- A D-W statistic below 2.0 and a p-value of 0.018 indicate significant positive autocorrelation, does not fulfill **independence** assumption.
- **Decision and Model Validity:** Although the independence assumption is not fully satisfied, the model still captures a strong biological trend and provides useful descriptive insights. However, the results should be interpreted with some caution.

Temperature Final Model (Linear Regression)

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \epsilon_i \\ &= 55.22 - 8.55 * (Temp) + 0.45 * (Temp^2) - 0.008 * (Temp^3) + \epsilon_i \end{aligned}$$

Where:

Y_i : Diameter of the Fungal Growth

X_1 : Temperature effect (linear term, $Temp$)

X_2 : Temperature effect (quadratic term, $Temp^2$)

X_3 : Temperature effect (cubed term, $Temp^3$)

ϵ_i : random error

Temperature Final Model (Linear Regression)

Predictor	Estimate	Std. Error	t value	p-value
(Intercept)	55.218	9.292	5.943	<0.001
Temp	-8.554	1.382	-6.191	<0.001
Temp ²	0.454	0.067	6.785	<0.001
Temp ³	-0.008	0.001	-7.164	<0.001

Table 4: Temperature Model Polynomial Regression Results

* Model based on formula: $\text{Diameter} \sim \text{Temp} + \text{Temp}^2 + \text{Temp}^3$

R²: 0.959 & Adjusted R²: 0.956:

- Excellent performance
- Explained 95% of variance
- Small gaps between R² and Adjusted R²
- Strong predictive power

F test p-value: < 2.2e-16

- Temperature is a strong factor to explain fungal growth

pH level Model

The background is a light gray gradient. It features several decorative elements: a large yellow shape in the top-left corner, a teal diamond in the middle-left, a yellow shape in the bottom-left, a yellow shape in the bottom-right, a thin blue circle outline on the right edge, and several small colored diamonds (orange, yellow, pink, blue) and thin blue circles scattered across the page.

ANOVA Table

Predictor	Df	Sum Sq	Mean Sq	F value	p-value
pH	1	0.0031	0.0031	2.3400	0.1520
pH ²	1	0.0118	0.0118	8.8743	< 0.05
Trial	1	0.0008	0.0008	0.5733	0.4636
Interactions					
pH:Trial	1	0.0080	0.0080	5.9981	< 0.05
pH ² :Trial	1	0.0030	0.0030	2.2391	0.1604
Residuals	12	0.0159	0.0013		

Table 5: ANOVA Table for pH Regression Model

Response Variable: Diameter

* Note: Cubic terms (pH³) omitted due to insufficient degrees of freedom.

* Significance level: 0.05

pH² (p=0.012): Confirms a strong curved/quadratic relationship between pH and fungal growth.

pH:Trial (p=0.03): The interaction terms with the Trial and pH is also significant to the model.

Conclusion: Keep the curvature term and the Trial factor. **Further investigation on term selection is required.**

pH Level Model Comparison using ANOVA

Model	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1: $pH \times Trial + pH^2$	13	0.018889				
2: $(pH + pH^2) \times Trial$	12	0.015919	1	0.0029702	2.2391	0.1604

Table 6: ANOVA Model Comparison for pH Interaction Complexity

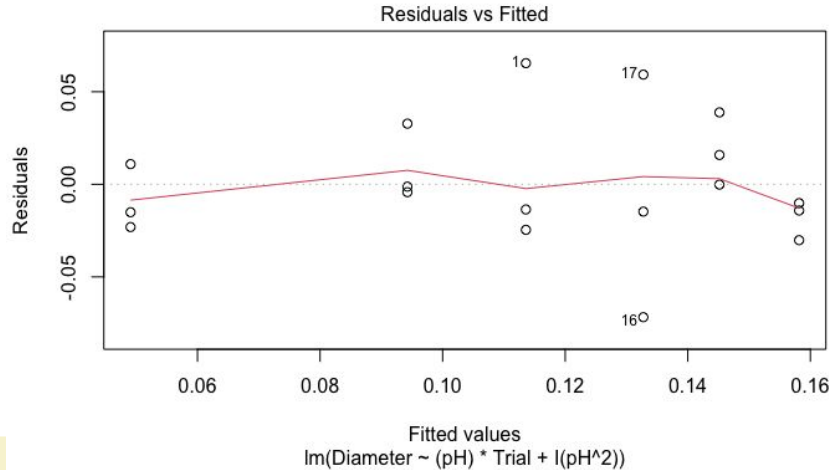
We compare the two models, with or without the interaction term (**pH² : Trial**) using ANOVA.

Result is **insignificant**, indicates dropping the interaction term does not significantly affect the model performance.

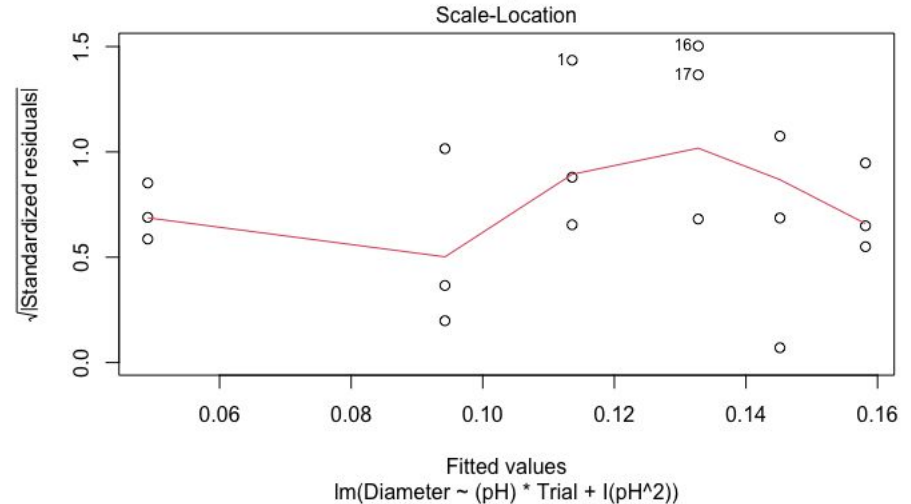
Thus, we proceed with the model without the interaction term for simplification.

Assumption Checks: Linearity, Homoscedasticity

Residuals vs Fitted: pH Model



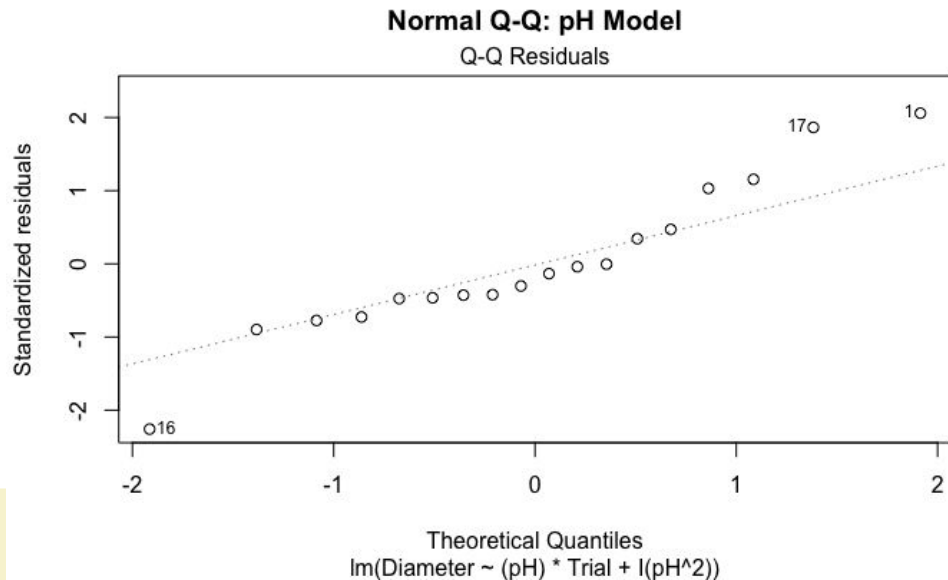
Scale-Location: pH Level Model



The Residuals vs Fitted plot shows a red smoothing line that stays relatively close to the horizontal zero line. The residuals are randomly scattered, indicating that the **linearity** assumption is satisfied.

The Scale-Location plot displays a slight peak in standardized residuals around the 0.13 fitted value, largely driven by observations 1, 16, and 17. However, the absence of a systematic "fanning" or "funneling" pattern suggests that **homoscedasticity** is sufficiently satisfied for this biological dataset.

Assumption Checks: Normality



Shapiro-Wilk normality test

$$W = 0.943$$
$$p\text{-value} = 0.326$$

The null hypothesis states that the residuals are normally distributed. With a p-value of 0.326, we fail to reject null hypothesis.

The Normal Q-Q plot shows the majority of standardized residuals following the diagonal reference line, with minor deviations at the upper tail that are typical in biological datasets. The Shapiro-Wilk test results fail to reject the null hypothesis, formally confirming that the residuals are normally distributed.

Assumption Checks: Independence

Durbin-Watson independence test

$$\begin{array}{|l} \text{D-W stat} = 2.50 \\ \text{p-value} = 0.954 \end{array}$$

- **P-value > 0.05:** We fail to reject the null hypothesis
- **Null Hypothesis (H_0):** There is no autocorrelation (residuals are independent).
- **Conclusion:** Since $p=0.954$ is very high, there is no evidence of autocorrelation. Therefore, the independence assumption is fulfilled.

pH Level Final Model (Linear Regression)

Predictor	Estimate	Std. Error	t value	p-value
(Intercept)	-0.452167	0.324937	-1.392	0.1874
pH	0.248556	0.104262	2.384	<0.05
Trial	-0.219000	0.089845	-2.438	<0.05
pH ²	-0.024111	0.008471	-2.846	<0.05
pH:Trial	0.034333	0.014672	2.340	<0.05

Table 7: Finalized pH Model Polynomial Regression Results

- * Residual standard error: 0.03812 on 13 degrees of freedom
- * Multiple R-squared: 0.5554, Adjusted R-squared: 0.4186
- * F-statistic: 4.059 on 4 and 13 DF, p-value: 0.02375

R²: 0.555 & Adjusted R²: 0.419

- Moderate performance
- Explained 55% of variance
- Reliable predictive power

pH² (p: 0.0138):

- Smallest p-value
- Proves fungal growth follows a non-linear, parabolic trajectory.

F test p-value: < 0.0238

- pH is a strong factor to explain fungal growth

pH Final Model (Linear Regression)

$$Y_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \epsilon_i$$

$$= -0.452 + 0.249*(pH) - 0.219*(Trial) - 0.024*(pH^2) + 0.034*(pH: Trial) + \epsilon_i$$

○

Where:

Y_i : Diameter of the Fungal Growth

X_1 : pH level effect (linear term, pH)

X_2 : Trial

X_3 : pH level effect (quadratic term, pH^2)

X_4 : pH:Trial interaction term

ϵ_i : random error

Conclusion

- **Temperature:**

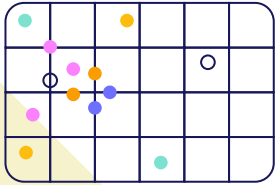
- Cubic model explains **95.6% of fungal growth variance**.
- Diameter is greatest at approximately **25°C** based on curve.
- Extreme temperature condition (0°C and 31.1°C) lead to **zero growth**.
- Temperature is a **more profound factor** than pH level on fungal growth.

- ● **pH Level Model:**

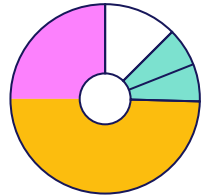
- Quadratic model identifies a significant **non-linear growth peak**.
- Optimal growth rate occurs at **pH 6**.
- Significant Trial interaction accounts for slight **experimental variation**.

- **Project Notes:**

- **Excluding zero-growth data** ensures statistically valid assumption testing.
- Moderate R^2 remains **scientifically useful** for biological inference.



Questions?



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